

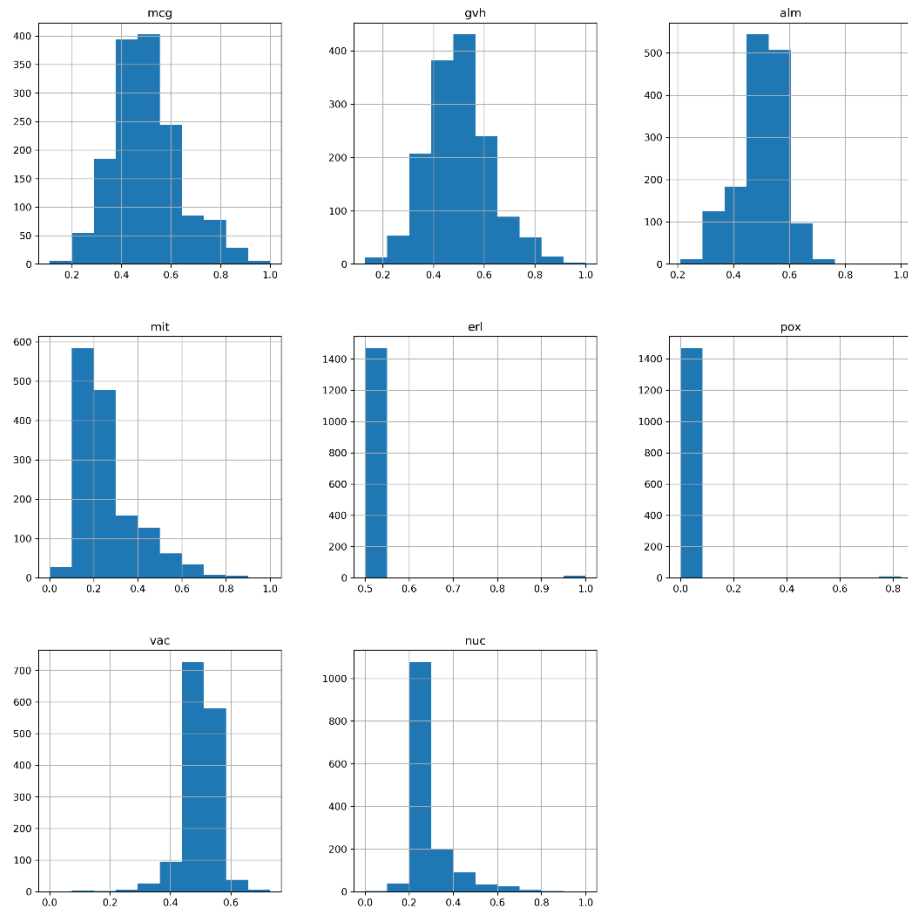


Figure S1: Histogram graphs of features.

Histogram Analysis

- mcg, gvh, and alm are relatively concentrated around the middle range, mostly between 0.3 and 0.7.
- Among them, mcg and gvh show a slightly right-skewed pattern, while alm is more centered and slightly left-skewed.
- mit is clearly right-skewed, with most proteins having low-to-moderate mitochondrial signal scores and fewer proteins showing high values.
- eri and pox are highly sparse. Most samples have the same low/default value, while only a very small number of proteins show high values.
 - This supports the idea that ER retention and peroxisomal targeting signals are rare in this dataset.
- vac is concentrated around relatively high values, especially near 0.5, with a longer tail toward lower values.
 - This explains its negative skewness.
- nuc is also right-skewed, with most values concentrated around 0.2–0.3 and fewer proteins having high nuclear localization scores.

- Overall, the histograms confirm that the features are normalized between 0 and 1, but their distributions are not uniform.
- Some features represent broadly distributed signals, while others capture rare biological signals.

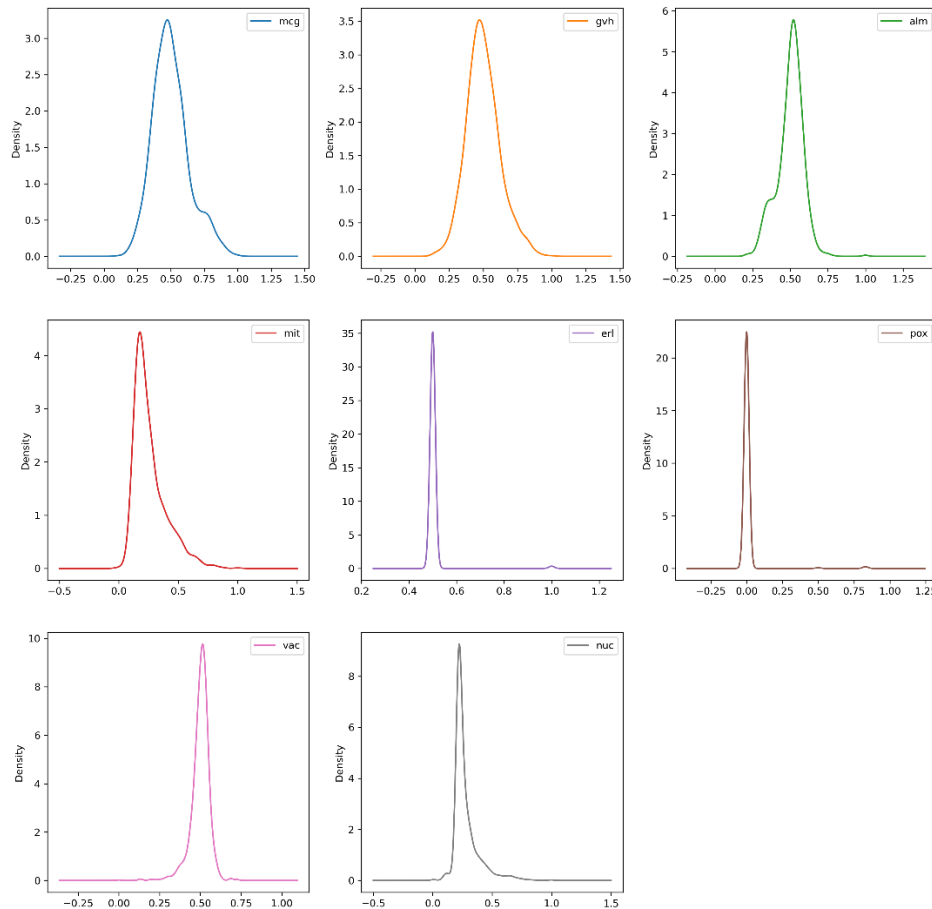


Figure S2: Density plot of features.

Density Plot Analysis

- mcg, gvh, and alm show approximately bell-shaped distributions centered around mid-range values, indicating that these signals are relatively equally distributed across proteins.
- mit is clearly right-skewed, with a strong peak at low values and a long tail toward higher values.
 - o This suggests that most proteins have low mitochondrial targeting scores, while fewer proteins show strong signals.
- erl shows an extremely sharp peak around ~0.5, indicating that most samples share nearly identical values.

- A very small secondary bump near higher values suggests rare activation of this signal.
- pox has a sharp peak near zero, with a long right tail.
 - o This confirms that most proteins lack this signal, while only a few exhibit strong peroxisomal targeting signals.
- vac is highly concentrated around ~0.5 with a slight tail toward lower values.
- nuc shows a strong peak at low-to-mid values with a long right tail, indicating that high nuclear localization scores are less frequent.

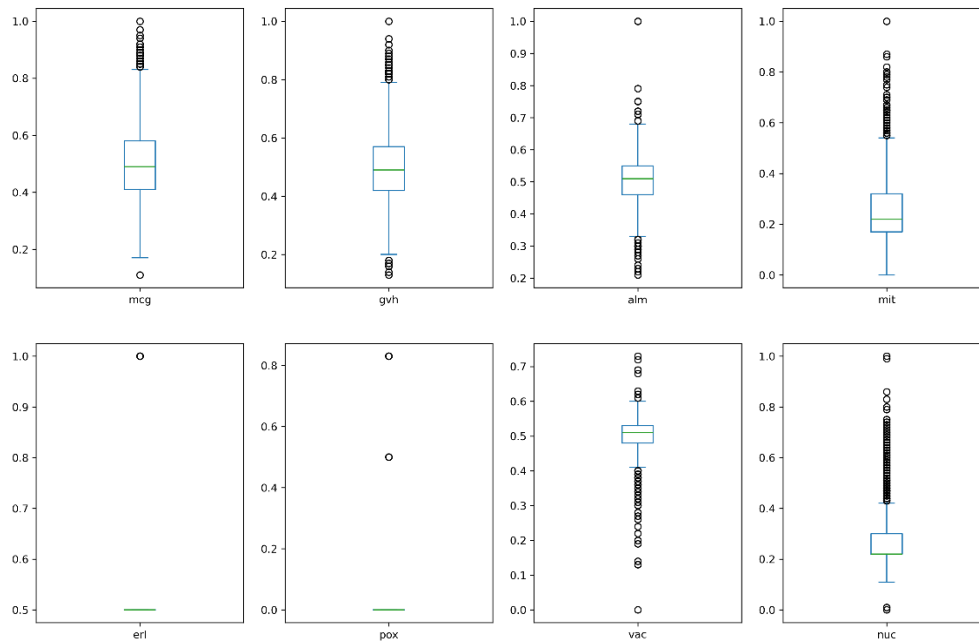


Figure S3: Box plot analysis of features.

Boxplot Analysis

- mcg, gvhl, and alm show relatively symmetric distributions with moderate spread.
 - o However, several upper (mcg, gvhl) and lower (alm) outliers are present, indicating proteins with unusually high and low signal scores.
- mit is clearly right-skewed, with the median located toward lower values and many outliers extending toward higher values.
 - o This confirms that strong mitochondrial signals are less common.
- erl has almost no variability: the box is extremely compressed around ~0.5, with only a few extreme outliers near 1.0.
 - o This indicates that most proteins share nearly identical values, with rare deviations.
- pox shows a similar pattern: most values are concentrated at the lower end (near 0), with a few high-value outliers.
- vac is tightly centered around ~0.5 but shows more spread compared to erl.

- It also contains both lower and upper outliers, with more variation toward lower values, consistent with negative skewness.
- nuc shows a strong right-skewed distribution, with many upper outliers.
 - This indicates that while most proteins have moderate nuclear localization scores, some have significantly higher values.
- Overall, boxplots confirm that many features contain outliers, which are likely biologically meaningful rather than noise.
- Features with many outliers may contain important biological signals useful for classification.

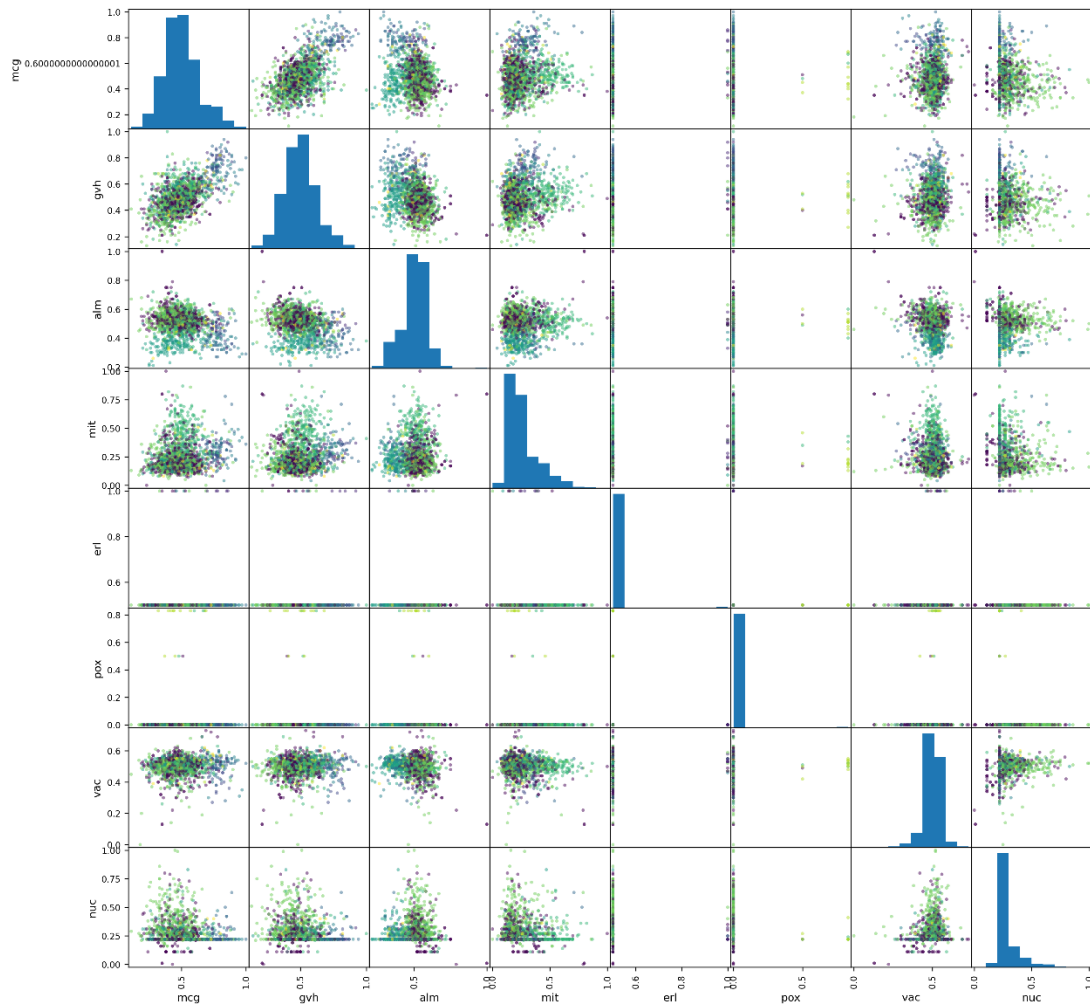


Figure S4: Scatter matrix analysis of the features.

Scatter Matrix Analysis

- The scatter matrix visualizes the pairwise relationships between all numerical features.
- Most feature pairs do not show clear linear relationships, consistent with the previously observed low correlation values.

- The only trend occurs between mcg and gvh, where a moderately positive relationship can be observed. This is consistent with correlation analysis.
- No clear separation appears between data points in any feature pair.
- The absence of apparent separation indicates that the dataset is not linearly separable.
 - This suggests that simple linear models may be insufficient, but nonlinear models are more appropriate.